Elasticity and Material Behavior

Key Definitions

- **Load:** The force applied to a material
- **Extension (x):** The increase in length of a material under a tensile force
- **Compression:** The decrease in length of a material under a compressive force
- **Limit of Proportionality:** The point beyond which extension is no longer proportional to the applied force (Hooke’ s Law no longer applies)
- **Elastic Deformation:** When the deforming force is removed, the material returns to its original shape and size
- **Plastic Deformation:** When the deforming force is removed, the material does not return to its original shape (permanent deformation)
- **Elastic Limit:** The point beyond which the material will not return to its original length and will be permanently deformed (very close to, or the same as, the limit of proportionality)

Hooke’ s Law and Spring Constant

- **Hooke’ s Law:** The extension of a material is directly proportional to the applied force, provided the limit of proportionality is not exceeded
- **Equation:** $F = kx$
- **Spring Constant (k):** $k = \frac{F}{x}$
  - Units: Newtons per metre ($\mathrm{N/m}$)
  - k is a measure of stiffness - higher k means a stiffer spring

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Testing Hooke’ s Law

Method: By Calculation (Using Ratios)

1. For a range of loads (F), measure the corresponding extension (x).
2. **Option A:** Calculate $k = F/x$ for different data pairs - if k is constant, Hooke’ s Law is obeyed.
   - Example: $10.4 \text{ N}/40 \text{ mm} = 0.26 \text{ N/mm}$ and $6.5 \text{ N}/25 \text{ mm} = 0.26 \text{ N/mm}$ ✅
3. **Option B:** Calculate ratio of two forces and ratio of their extensions - if ratios are equal, Hooke’ s Law is obeyed.
   - Example: $5.2 \text{ N}/10.4 \text{ N} = 0.5$ and $20 \text{ mm}/40 \text{ mm} = 0.5$ ✅

Method: By Graph (Best Method)

1. Plot Force (F) against Extension (x).
2. If the graph is a straight line that passes through the origin, Hooke’ s Law is obeyed.
3. Mathematical Proof: Use $F = kx + c$. If y-intercept $c = 0$, the line passes through origin.

Experimental Determination of Elastic Limit

1. Set up apparatus (spring or wire).
2. Add small mass and measure extension.
3. Remove mass and check if spring returns to original length.
4. Repeat with increasingly larger masses.
5. **Elastic limit** = maximum load for which spring still returns to original length when completely removed.
6. **Graphically:** On Force-Extension graph, the point where graph first deviates from straight line.

Stress, Strain, and Young Modulus

- **Stress (σ):** $\text{stress} = \frac{F}{A}$ | Units: Pascals (Pa) or N/m^2
- **Strain (ε):** $\text{strain} = \frac{x}{l}$ | Units: None (ratio)
- **Young Modulus (E):** $E = \frac{\text{stress}}{\text{strain}} = \frac{Fl}{Ax}$ | Units: Pascals (Pa)
  - High Young Modulus = stiff material (difficult to deform)

Springs in Series and Parallel

Series Combination:

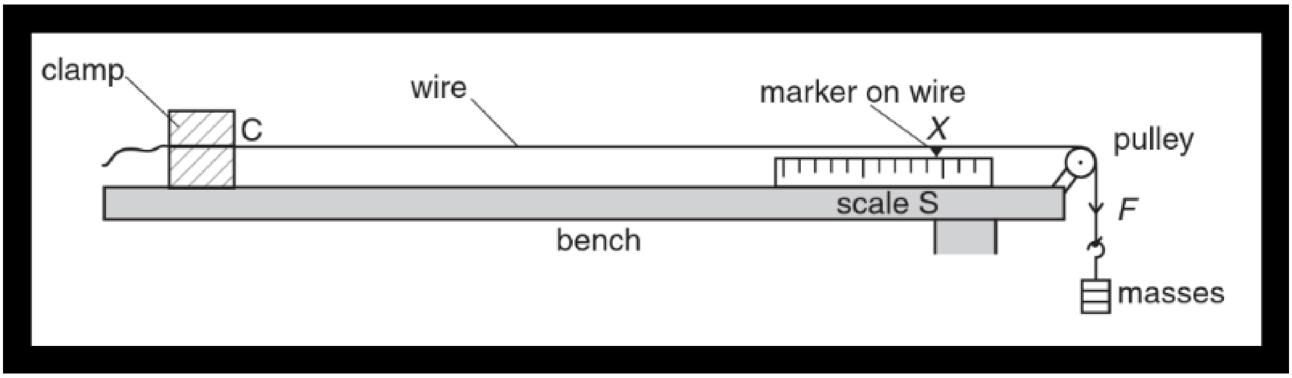
- Same force F through each spring.
- Total extension: $x_{\text{total}} = x_1 + x_2$
- Effective spring constant: $\frac{1}{k_{\text{series}}} = \frac{1}{k_1} + \frac{1}{k_2}$
- Combined spring is less stiff.

Parallel Combination:

- Load shared between springs: $F_{\text{total}} = F_1 + F_2$
- Same extension x for each spring.
- Effective spring constant: $k_{\text{parallel}} = k_1 + k_2$
- Combined spring is more stiff.

Young Modulus Experiment

Apparatus Diagram:



Procedure:

1. Set up long, thin wire clamped horizontally over pulley (or vertically from ceiling).
2. Add reference mark ⓒ on wire to define test section.
3. Fix scale (ruler) near mark to measure extension.
4. Measure original length (l) from clamp to reference mark.
5. Use micrometer to measure diameter (d) at several points $\rightarrow$ calculate average diameter.
6. Calculate cross-sectional area: $A = \frac{1}{4}\pi d^2$.
7. Add known mass (load, F) and record new position on scale.
8. Calculate extension: $e = \text{final reading} - \text{initial reading}$.
9. Repeat for range of loads.

Increasing Accuracy:

- Measure diameter in several places and take average.
- Use long, thin wire to make extension larger.

- Take several readings with different loads and plot graph.
- Remove load and check wire returns to original length (confirms testing within elastic limit).

Data Analysis for Young Modulus

Method: From Force-Extension Graph (Best)

1. Plot Force (F) against Extension (e).
2. Draw line of best fit through linear portion.
3. Gradient = spring constant k (where $F = ke$).
4. Calculate: $E = \frac{\text{gradient} \times l}{A}$

Elastic Potential Energy

- Area under Force-Extension graph = work done.
- For material obeying Hooke's Law (within limit of proportionality):
  - $E_p = \frac{1}{2}Fx$
  - $E_p = \frac{1}{2}kx^2$

Unloading Behavior

When you remove the load (deforming force) from a wire or spring, its behavior depends on whether it was deformed elastically or plastically:

Elastic Unloading (Within Elastic Limit)

- The wire/spring returns exactly to its original length.

- The Force-Extension graph shows a straight line back to the origin.
- All stored elastic potential energy is recovered.
- Molecular explanation: Atomic bonds were stretched but not permanently rearranged - they return to their original positions.

Plastic Unloading (Beyond Elastic Limit)

- The wire/spring returns partially but shows permanent extension.
- The Force-Extension graph shows the unloading line is parallel to the loading line but doesn’t return to origin.
- Molecular explanation: Atomic bonds have slipped past each other and cannot return to original positions.

Force-Extension Graph Showing Unloading